

THE LOG-NORMAL FUNCTION AS A STOCHASTIC MODEL OF THE DISTRIBUTION OF STRONTIUM-90 AND OTHER FISSION PRODUCTS IN HUMANS*

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Abstract—Data on the concentrations of ^{90}Sr , ^{226}Ra , ^{137}Cs , and some of the stable trace elements in human tissue as well as environmental samples have been found, without exception, to follow log-normal distributions rather than normal distributions. Also, while the median of the ^{90}Sr concentration in humans shifts for different time periods, areas of the world, and population age distributions, the geometric standard deviation has seemed recently to remain within the relatively narrow range of 1.55–1.67 for large population groups, so that about the same proportion of each population contains concentrations in excess of a given multiple of the median (or the average) concentration. The log-normal distribution fits the data better than the normal distribution, particularly in the “tail” of higher concentrations. This is the region most critical for the evaluation of the proportions of the population exceeding recommended limits of individual exposure. The repeated suitability of the log-normal permits the recommended limits of environmental contamination to be established with more confidence in predicting the proportions of the population exceeding any given level. The 1964–1965 U.S. Public Health Service measurements of ^{90}Sr in the bones of U.S. children and young adults give parallel cumulative log-normal lines for the ^{90}Sr concentration per gram of bone and per gram of calcium. These lines have approximately the same slope as Kulp and Schulert’s distribution for the variations in ^{90}Sr concentration in total skeletal bone between adults in New York City in 1958, and the same slope as the distribution for stable strontium, indicating a characteristic variance in the log-normal distribution of strontium in bone within a limited geographic region. The average of the United States distribution in children and young adults was about $3.5 \mu\text{Ci } ^{90}\text{Sr/g}$ calcium in October, 1964–March, 1965, the equilibrium level of $3.5 \mu\text{Ci/g}$ predicted by Kulp and Schulert for 4–5 yr following the 1961 Soviet tests. The log-normal line fitted to the 1964–1965 U.S. data indicates that less than 3.6% of the United States population of age less than 25 will have bone doses exceeding twice the average. Extrapolation of the log-normal distribution would indicate that less than 0.36% of the population under age 25 would exceed $10.4 \mu\text{Ci } ^{90}\text{Sr/g}$ calcium (3 times the average). The same percentages would apply to the corresponding multiples of the average ^{90}Sr concentration per gram of bone. The applicability of the log-normal function for estimating internal, and possibly external, exposure distributions in the world population is also discussed.

* This paper is dedicated to Theodore F. Hatch on the occasion of his attaining the rank of emeritus professor of industrial health engineering in the Graduate School of Public Health and in acknowledgement of his pioneering developments in the application of the log-normal distribution beginning in the 1920’s.

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I. INTRODUCTION

THE DISSEMINATION of ^{90}Sr and other fission products from nuclear bomb tests is world-wide. Further additions of fission products to man's environment result from the radioactive materials produced by atomic reactors. The possible future dispersal of fission products from nuclear-powered rockets upon re-entry from outer space must also be considered. If the quantities of fission products injected into the biosphere increase more rapidly than compensated for by radioactive decay and diffusion from the biosphere, the radiation exposure from these man-made nuclides may eventually exceed that from natural causes, or may exceed the recommended limits⁽¹⁾ for radiation exposure from man-made sources. Thus, it is necessary to periodically assess the radioactivity present in the human environment.

Many biological studies have led to a general recognition of the hazard and the establishment of recommended upper limits to the population exposure from man-made sources.⁽²⁾ The setting of these limits must be based on the following criteria:

- (1) A characteristic sample of the population to which some maximum permissible level (MPL) is assumed to apply;
- (2) A statistical model that describes the extent to which members of the sample will vary in their critical organ concentrations of the particular radionuclides of concern; and
- (3) A safety factor that seems reasonable in the light of the range of exposure over which the radiobiological phenomena must be extrapolated in order to predict low-incidence effects accumulated over large and heterogeneous populations.

This paper further illustrates the applicability of the log-normal distribution recommended⁽³⁻⁵⁾ for the second criteria listed above, when dealing with various samples of the population. In many investigations of the distribution of fission products in humans, the population under study has of necessity been confined to the people living in a given area or belonging to a given age-group. An MPL based on statistics derived from such samples can be misleading unless the relative sizes of each geographic or age-group sub-population, as well as their individual

exposure distributions, are taken into account in estimating the distribution of exposure to the world population. Thus, we wish to discuss also the extent to which the log-normal model may now be applicable to the radionuclide concentrations in various segments of the world population. Although the amount of data within homogeneous population sub-groups is still somewhat limited, certain trends may now be more apparent when the data are examined on a log-normal scale. Other skewed distribution models are not ruled out; however, they do not have all of the convenient properties of the log-normal function, as described in the next section.

In assessing the potential hazard of a radioisotope and determining the MPL, a critical consideration is the extent to which an appreciable segment of the global population accumulates 2, 3 or more times the average body burden.⁽³⁻⁵⁾ This point has been made by DAHL⁽³⁾ who wrote "... it is quite possible that even if the population as a whole has a concentration which represents no danger, a fairly large number of individuals may nevertheless have concentrations in the dangerous range." NEWMAN⁽⁴⁾ has, in a similar vein, written "... obviously, the average level (Sr-90) must be kept below the MPL by a factor which will allow for variability ... choosing this safety factor is quite analogous to choosing the MPL in the first place ..."

In this communication we shall use previously published data to further demonstrate that the frequency distribution of concentrations between humans of the world population—for strontium-90, radium, cesium and stable trace elements including strontium—is closely approximated in shape by the log-normal distribution, and that this distribution continues to characterize empirical information on isotopic distribution much better than the normal distribution, particularly for the higher concentration region that is most critical in the evaluation of the size of the population that may be exposed to the higher multiples of the average dose. Available data also indicate that the amounts of fission product activity deposited in various geographical areas may also approximate a log-normal variation in the external exposures contributed by fallout to the U.S. population.

II. THE LOG-NORMAL DISTRIBUTION

The log-normal distribution has been used to summarize and describe empirical data for many years, and many theses in biology, geochemistry, economics and other fields have been devoted to expounding its usefulness.⁽⁶⁻¹²⁾ DAHL⁽³⁾ and others^(5,13,14) have already applied the log-normal distribution to the problem of predicting the levels of ⁹⁰Sr in various proportions of the human population. The variation in the amounts of trace elements found in geological materials is also known to be closely approximated by the log-normal.⁽⁹⁾ However, limited use has been made of the log-normal so far in evaluating the causes and consequences of variations in radionuclide concentrations between humans.⁽¹⁵⁾ Also, incorrect definitions and applications of the log-normal have appeared in the scientific literature, as pointed out by ESPENSCHIED *et al.*⁽¹⁰⁾ Accordingly, an abbreviated review of the log-normal distribution and definitions of its parameters is given in this section before proceeding with further illustrations of its applicability.

A random variable X whose logarithm obeys the normal law of errors is said to be distributed according to the "log-normal distribution".⁽⁹⁾ The use of the log-normal distribution is convenient where applicable, since the normal theory of estimation and regression analysis⁽¹²⁾ is then applicable to the logarithms of the observed values of x . The distribution function of the log-normal is given by:

$$P(\ln X \leq \ln x) = \frac{1}{\sqrt{2\pi} \sigma_g} \int_{\ln X = \ln 0 = -\infty}^{\ln X = \ln x} \times \exp \left[-(\ln X - \ln \mu_g)^2 / 2\sigma_g^2 \right] d(\ln X) \quad (1)$$

where $P(\ln X \leq \ln x)$ is the probability that a value of $\ln X$ will be less than or equal to $\ln x$, σ_g is the standard deviation ($\sqrt{\text{Var}(\ln X)}$) in the frequency distribution of $\ln X$, and $\ln \mu_g$ is the mean value of $\ln X$, μ_g being defined in this paper as the "geometric mean of X ". The expression in equation (1) is the well-known form for the cumulative normal distribution for $\ln X$, and has recently been shown to be a special case of a family of logarithmic distribution functions.⁽¹⁰⁾ Since the log-normal distribution has been found to give an adequate fit to the bone data, and since the log-normal

is more convenient to use as discussed above, no other skewed functions are considered here as a model for radionuclide distributions in the human population. However, other skew distributions may later warrant consideration if future data, particularly in the extremes of the frequency scale, indicate the need for additional models.

According to equation (1) and the method of constructing commercial probability graph paper, a plot of the concentration x on the log scale versus the percent of the population with concentrations $\leq x$ on the percentage scale of probability vs. log graph paper will yield a straight line when the distribution of X is log-normal. The value of x where the line crosses the 50 percentile is μ_g , the ("geometric") mean and also the median concentration for the log-normal distribution. The value of $\ln \mu_g$ is the mean value of $\ln X$. Since $\log_{10} x = \ln x / \ln 10 = \ln x / 2.303$, $\log_{10} x$ plotted vs. percent with concentrations $\leq x$ will also yield a straight line on probability vs. log coordinate paper. Similarly, $\log_{10} \mu_g$ will be the average value of $\log_{10} X$.

The cumulative distribution in X rather than $\ln X$ is obtained directly from equation (1) by substituting for $d(\ln X) = dX/X$, and changing limits appropriately:

$$P(X \leq x) = \frac{1}{\sqrt{2\pi} \sigma_g} \int_{X=0}^{X=x} \times \frac{1}{X} \exp \left[-\frac{1}{2\sigma_g^2} (\ln X - \ln \mu_g)^2 \right] dX. \quad (2)$$

The integrals are, of course, defined as the limits of the corresponding integrals over the domain (E, x) as $E \rightarrow 0$, for $0 < E < x$. From equation (2), the population mean (or "average") value* of X is

$$\bar{x} = \frac{1}{\sqrt{2\pi} \sigma_g} \times \int_{X=0}^{X=\infty} \exp \left[-\frac{1}{2\sigma_g^2} (\ln X - \ln \mu_g)^2 \right] dX \quad (3)$$

* The population mean in the statistical literature is also called the "expected value" or $E(X)$. In this paper, the symbol $\bar{x}$ is used to designate $E(X)$ rather than the average of a sample of values of x : Other symbols are also defined in such a way to make this summary of the log-normal distribution more easily understood by the non-statistician.

since the X in the denominator of the integrand in equation (2) cancels the X being averaged over the interval $(0, \infty)$.

The average value $\bar{x}$ can be shown⁽¹⁰⁾ to be related to μ_g and σ_g by the equation

$$\ln \bar{x} = \ln \mu_g + \sigma_g^2/2 \quad (4)$$

where $\ln \mu_g$ is the mean value of $\ln X$ as noted above, and σ_g is defined as the standard deviation in $\ln X$ such that

$$\sigma_g^2 = \frac{1}{\sqrt{2\pi} \sigma_g} \int_{\ln X = -\infty}^{\ln X = \infty} (\ln X - \ln \mu_g)^2 \times \exp \left[-\frac{1}{2\sigma_g^2} (\ln X - \ln \mu_g)^2 \right] d(\ln X). \quad (5)$$

Now, one of the major conveniences in the use of the log-normal rather than other logarithmic distributions is the fact that estimates of μ_g and σ_g may be easily obtained from a plot of percent of population with concentrations $\leq x$ versus the value of x itself, as indicated on probability vs. log scale graph paper. The value of μ_g is simply the value of x at which the plot crosses the 50% population line. The estimated value $\hat{\sigma}_g$ of σ_g is obtained from

$$\hat{\sigma}_g = \ln x_{0.8413} - \ln x_{0.50} = \ln (x_{0.8413}/x_{0.50}) \quad (6)$$

or

$$\hat{\sigma}_g = \ln x_{0.50} - \ln x_{0.1587} = \ln (x_{0.50}/x_{0.1587})$$

where $x_{0.8413}$ is the value of x where the “% population vs. log X ” line crosses the 84.13% population value, and $x_{0.50} = \mu_g$ is the value of x where the line crosses the 50% population value. The estimated value of $\bar{x}$ may then be easily calculated from μ_g and $\hat{\sigma}_g$ by equation (4). The various log-normal distributions found to fit the variations in element and radionuclide concentrations in human populations will be compared using equations (1)–(6).

Defining the “standard geometric deviation”⁽⁷⁾ or the “geometric mean standard deviation”⁽¹⁰⁾ to be

$$s_g = x_{0.8413}/x_{0.50} = x_{0.50}/x_{0.1587}, \quad (7)$$

and converting to common logarithms using the relation $\ln x = 2.303 \log_{10} x$, equation (4) is changed to the form used by HATCH⁽⁷⁾

$$\log_{10} \bar{x} = \log_{10} \mu_g + 1.1513(\log_{10} s_g)^2. \quad (8)$$

Equations (1), (2), (3), (5) and (6) may also be easily expressed in terms of common logarithms in obvious fashion.

The probability density function $dP/d(\ln X)$ obtained by differentiating equation (1) would appear as the symmetric bell-shaped “normal” or “Gaussian” curve when plotted on a linear frequency scale versus the logarithm of X (or the values x plotted on a logarithmic scale). In this case, the mode or peak of the distribution curve would fall at the expected (mean) value of $\ln X$, which would also be the median value of $\ln X$ and would, of course, be the same as the natural logarithm of the median value μ_g of X .

The probability density function dP/dX , on the other hand, would appear as a distribution skewed to the right, identically equal to zero for $x \leq 0$, when dP/dX is plotted on a linear frequency scale versus values of X plotted on a linear scale. The mode of this distribution curve would be to the left of the median value μ_g of X and the expected value of X would lie to the right of μ_g by an amount dependent on σ_g , as given in equation (4).

The maximum likelihood estimate of μ_g for a population having a log-normal distribution in ^{90}Sr per gram calcium with parameters μ_g and σ_g is the geometric mean μ_g of the concentrations $x_1, x_2, \dots, x_n$ of a “random” sample representative of individuals from the population (selected so that any member of the population has an equal chance of falling into the sample). The value of μ_g is given by

$$\hat{\mu}_g = (x_1 x_2 \dots x_n)^{1/n}. \quad (9)$$

The maximum likelihood estimate (corrected for bias) of T_g may also be calculated from the equation

$$\hat{\sigma}_g = \left(\left[\sum_{i=1}^n (\ln x_i - \ln \mu_g)^2 \right] / [n - 1] \right)^{1/2} \quad (10)$$

Substitution of equations (9) and (10) into equation (4) would yield the estimate of the mean (or average) concentration $\bar{x}$. However, additional uncertainties will occur in the estimation of mean radionuclide concentrations from existing data to the extent that the population samples are not truly representative. Thus, for purposes of this paper, it will usually suffice to obtain the estimates of μ_g and s_g graphically

as described by equation (7) and the preceding paragraphs.

The standard deviation σ_x of the frequency distribution of x , defined by

$$\sigma_x^2 = \frac{1}{\sqrt{2\pi} \sigma_g} \int_{x=0}^{x=\infty} (X - \bar{X})^2 \frac{1}{X} \times \exp [-(\ln X - \ln \mu_g)^2 / 2\sigma_g^2] dX \quad (11)$$

may be obtained from μ_g and σ_g^2 by the equation⁽¹⁷⁾

$$\sigma_x^2 = \exp (2 \ln \mu_g + 2\sigma_g^2) - \exp (2 \ln \mu_g + \sigma_g^2). \quad (12)$$

The value of σ_x is not as useful for log-normally distributed variables as s_g or σ_g , since $\hat{\mu} \pm \frac{x}{s_g}$ and not $\hat{x} \pm \sigma_x$ gives the 68% confidence intervals for the estimation of the "true" value of x when the frequency distribution of $\ln x$ is presumed to be a normal "error" distribution.

The range $\hat{\mu}_g \pm \frac{x}{s_g}$ would be simply an estimated interval covering 68% of the population when the variations in x are not considered merely errors in its determination. Similarly, $\hat{\mu}_g \pm \frac{x}{s_g^2}$ rather than $\hat{x} \pm 2\sigma_x$ would approximate the "95% confidence" interval.

The repeated empirical demonstration that the log-normal distribution adequately describes the variations of radionuclide concentrations in various human sub-populations would suffice to justify its use merely as a tool for increasing the reliability and ease of estimating average population exposures, and/or proportions of the population exceeding various multiples of the average. Nevertheless, it would be desirable to determine the theoretical basis of the log-normal distribution of radionuclide concentrations so that we may be alerted to its limitations and the nature of the conditions under which it might not be applicable.

Although FINNEY⁽¹²⁾ states that the probit (or log-normal) transformation is only a convenient trick for representing the sigmoid curves in many bioassay experiments by a straight line, GADDUM⁽⁹⁾ presents a number of heuristic arguments for the applicability of the log-normal rather than the normal function as a description of the frequency distributions of various kinds

of data. He notes the fact that the log-normal meets the boundary condition $P(X \leq 0) = 0$ that applies to any variable that cannot take on negative values. He also points out that a log-normal distribution in an observable quantity X should be expected when the value of $\ln X$ is the result of a large number of independent causes, each randomly distributed and adding a small contribution to the value of $\ln X$.

Suppose the value of X could be considered to be the product of N factors,

$$X = K \prod_{i=1}^N y_i^{n_i} \quad (13)$$

where the y_i are random variables and the n_i are constants. By taking the logarithm of both sides of equation (13) we have

$$\ln X = \ln K + \sum_{i=1}^N n_i \ln y_i. \quad (14)$$

From equation (14), we can easily see that if each of the variables y_i is log-normally distributed then X must be log-normally distributed, even if N is not large. Under more general conditions, if the y_i are independently distributed then equation (14) would (by the Central Limit Theorem) approach normality as the number N of variables y_i becomes larger.⁽¹⁸⁾ A sufficient condition (Liapounoff condition^(18,19)) for the validity of the Central Limit Theorem essentially requires that each random variable in the summation of equation (14) should contribute, on the average, only a small amount to the summation.

As for many other mathematical relationships between measurable quantities, there are reasons for expecting that the quantity of ^{90}Sr in the bone of an individual is the product of a number of multiplicative factors, as in equation (13). Some factors already identified are the deposited concentrations in soil or on plants, the fractional uptake from soil into plants relative to calcium, the fractional uptake from plants into milk relative to calcium, and the fractional uptake from milk or other foods into human bone.⁽¹³⁻¹⁵⁾ Not only would we expect these factors to be multiplicative in determining the quantities of ^{90}Sr in human bone, but we may also expect each or all of them to be log-normally distributed, or approximately so. First, none of

these factors can take on negative values and some of them may be expected to vary over a wide range of values from fractions close to zero to factors greater than one. Thus, the suitable boundary condition $P(X \leq 0) = 0$ for the log-normal model near the value $x = 0$ may be expected to favor it over the normal distribution. Further, each of the uptake factors is itself the resultant of random processes consisting of a large number of events, where each event may tend to increase or decrease the uptake by a small proportion.

Also, any variable $X = \exp Y$ is log-normally distributed when Y is normally distributed.⁽¹⁶⁾ Various kinds of attenuation factors have such a form,⁽²²⁾ and Y will be approximately normally distributed if not exactly, whenever its value is proportional to the sum of the number of successes in a large number of binomial trials⁽²³⁾ (e.g. atoms with a specific probability p of penetrating some barrier). Of course, since $\exp Y = 1 + Y + Y^2/2 + \dots$, the log-normal and normal distributions will practically coincide over most of the probability space when the coefficient of variation in Y is less than about 0.14.⁽⁹⁾

The following section describes some of the recent evidence of log-normal variations in uptake factors as well as in human bone concentrations of ^{90}Sr and other nuclides. Much of the following discussion is presented more to

illustrate how the slopes and intercepts of cumulative log-normal frequency data, transformed into straight lines on a simple probit-type plot, may be inter-compared and utilized for studying the degree and sources of variability rather than to derive any final conclusions regarding the mechanisms of fallout transport to humans. In this way, the authors hope to encourage more frequent use of the log-normal probability plot, which requires only simple arithmetic calculations and plotting.

III. THE LOG-NORMAL DISTRIBUTION FITTED TO DATA ON NUCLIDE CONCENTRATIONS IN HUMAN POPULATIONS AND THEIR ENVIRONMENTS

Natural Element Concentrations in Humans

DAHL⁽³⁾ showed that the distribution of the natural non-radioactive strontium to calcium ratio in the world population, as measured by Turekian and Kulp, was represented by a log-normal function. A plot of this distribution is shown in Fig. 1. Also plotted in Fig. 1 for comparison is the normal approximation to the histogram, illustrating the relative shapes of the two distributions on a rectangular coordinate scale; and the better fit of the log-normal distribution to the data. Although the 277 bone specimens obtained from various

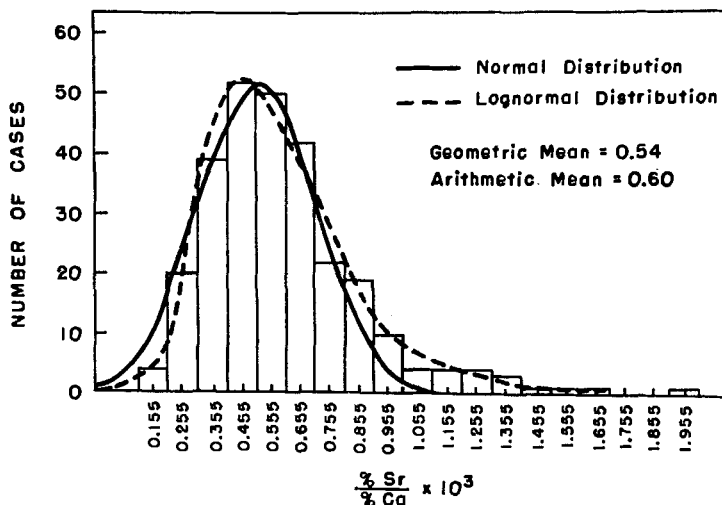


FIG. 1. Comparison of shapes of normal and log-normal frequency distributions on rectangular coordinates scales, from Ref. 5 (as replotted from the original data of Ref. 24).

areas of the world were not necessarily obtained by sampling procedures that appropriately weighted the relative population sizes in the different regions, the data nevertheless fits a log-normal function over a relatively wide concentration range.

A quick method for judging whether a population is log-normal or normal involves a comparison of cumulative frequency plots on both arithmetic and logarithmic probability paper (Ref. 11, pp. 31–33). The use of logarithmic or arithmetic probability paper serves as a convenient graphical method of analysis for evaluating the applicability of the normal or log-normal distributions (see Fig. 2).

In Fig. 3, we have replotted the distributions of several other naturally-occurring elements in United States populations, as reported by SNYDER and COOK.⁽²⁵⁾ All of these distributions plotted on coordinates that linearize the cumulative log-normal distribution, are seen to be fitted very well by the log-normal model—well within the confidence limits for the data as given by SNYDER and COOK.⁽²⁵⁾ * It may be of

interest to note in Fig. 3 that elements having similar valence properties have similar degrees of variation s_g .

Radionuclide Concentrations in Humans

DAHL⁽³⁾ also demonstrated that plots of the distribution of the $^{90}\text{Sr}/\text{Ca}$ ratio in human bones from North America,⁽²⁶⁾ and the distribution in the $^{90}\text{Sr}/\text{Ca}$ ratio in fifty-three stillborn children in Chicago in 1953,⁽²⁷⁾ also fitted the log-normal. Further, the distribution in stillborn children was found to have about the same s_g as the distributions of natural strontium and radium in human skeletons. The $^{90}\text{Sr}/\text{Ca}$ distribution of KULP, ECKELMANN and SCHULERT⁽²⁶⁾ was much wider (larger s_g) and originally caused considerable concern about the appreciable fraction of the world population that might have ^{90}Sr concentrations many times the world average.^(3–5)

In Fig. 4 we have plotted some of the $^{90}\text{Sr}/\text{Ca}$ distributions from different population groups over the period 1956–1965. All of the distributions are found to be more consistent with a

* Since the writing of this paper a computer program has been made available to us through the courtesy of W. S. Snyder and Mary Jane Cook of the Oak Ridge National Laboratory for the testing of hypotheses of normality or log-normality using three independent statistics calculated from the sample of data under analysis. These tests were programmed by Barbara Ann Flores and Monica Boyd formerly of Oak Ridge National Laboratory, and are based on tests of mean deviation/standard deviation, skewness, or kurtosis for consistency with the shape of the normal or log-normal distribution. These tests are given in K. PEARSON and H. O. HARTLEY (*Biometrika Tables for Statisticians*, Vol. 1, pp. 61–62, published by Cambridge University Press, 1954). The data of Fig. 4 for ^{90}Sr in Denmark foetuses, (from a sample of thirty-one values) and the data on Fig. 6 for ^{137}Cs in milk (from a sample sixty-three values), have shown much better agreement with the hypothesis of log-normality than with the hypothesis of normality for each of the three sample statistics. Only these two sets of data have been tested using the computer program. However, it is evident from the goodness of fit to the log-normal, as visually observed for the other data in Figs. 3–6, and the fact that the s_g values and degrees of freedom of most of the other data are at least as large as those of the plot for ^{90}Sr in Denmark foetuses, that the other data would be expected to be in much

better agreement with an hypothesis of log-normality than an hypothesis of normality, and would be consistent with the hypothesis of log-normality at the $P = 0.90$ level or better. The authors wish to emphasize that a visual choice between the log-normal and normal hypotheses may usually be made quite easily by the simple arithmetic calculation of the fractional amounts of data below each of a set of concentration values, followed by plotting on both normal and log-normal probability paper. Such a comparison is illustrated for the early bone data⁽²⁹⁾ of Fig. 4 replotted in Fig. 2 to the same ordinate scales with both linear and logarithmic abscissae. Of course, the other data of this paper with smaller s_g values would show a less striking difference between the two plots, but the better fit to a straight line would be clear.

Nevertheless, when a hard and objective choice (and particularly many hard choices) must be made between the log-normal and normal or other distribution representations, computer-programmed statistical tests are useful and convenient to use.

For field tests and empirical comparisons of variability, the graphical plot usually suffices; and when the choice is difficult in these cases, the distribution is so close to either normality or log-normality that the choice of coordinates makes little difference for the purpose at hand.

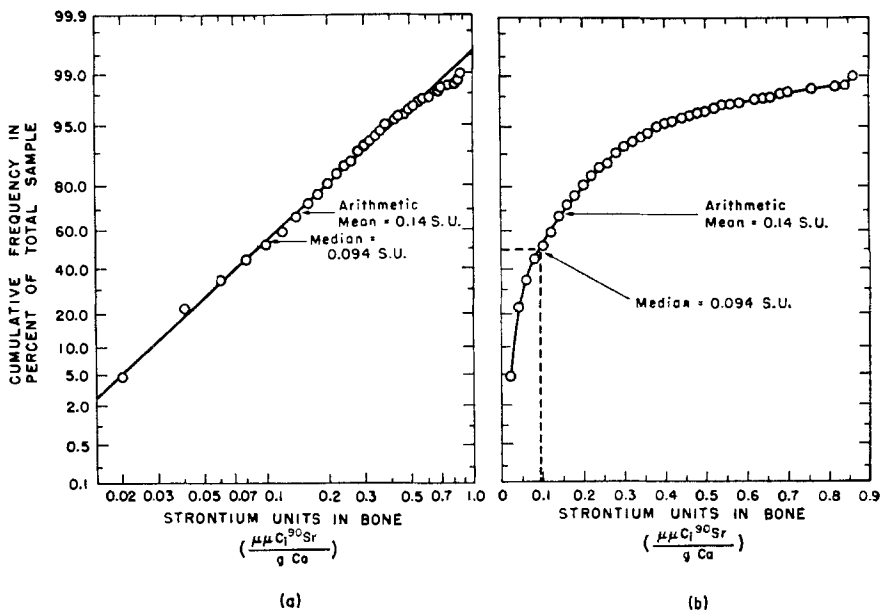


FIG. 2. Comparison on log-normal (a) and normal (b) coordinates of cumulative frequency plots of the data⁽²⁹⁾ on concentrations of $^{90}\text{Sr}/\text{Ca}$ in the bones of 838 adults from throughout the world in 1956–1958 as previously illustrated from Ref. 5. The straight line in Fig. 2 (a) contrasted with the curve in Fig. 2 (b) shows very clearly in this case that the distribution is very close to log-normal and inconsistent with any hypothesis of normality.

log-normal than a normal model, as found with the earlier ^{90}Sr data.^{(3–5)*} The data for children in Denmark in 1961 includes only seventeen individuals, so deviations from the log-normal line are well within ordinary sampling variations. Figure 4 shows that data from more limited geographical areas give log-normal lines that are steeper (representing less variation) than the log-normal lines of KULP, ECKELMANN and SCHULERT⁽²⁶⁾ for North America in 1955. Even the measurements of KULP, SCHULERT and HODGES⁽²⁹⁾ on 838 adult human bone samples from different regions of the world for the period 1 July 1956 to 30 June 1958, plotted in Fig. 4, gives a smaller range of variation than the earlier data for North America alone. This decrease in the variation between individuals is probably not due only to improvements in the precision of radiochemical measurements,⁽³⁰⁾ but rather to a smoothing of the variability of ^{90}Sr deposition and distribution in food with time during the first few years following the early hydrogen bomb tests.

In Fig. 4, Kulp and Schulert's distribution

for whole skeleton adult bone samples⁽¹⁴⁾ obtained in New York City over the 1956–1959 period (but normalized to the 1958 average level) is replotted for comparison on the same scale with the distributions⁽²⁸⁾ in fetuses and in children in Denmark in 1961, and with children and young adults in the Northeastern section of the United States in 1964–1965.⁽³¹⁾ The slopes of all these $^{90}\text{Sr}/\text{Ca}$ distributions after 1958 are almost the same, and the slopes for the 1961–1962 and 1964–1965 lines are almost identical with that of the line for the natural strontium/calcium ratio given in Fig. 3. Thus, the degree of individual variation shown by these distributions may be indicative of a characteristic variability between individuals of the same region and age-group who have been ingesting ^{90}Sr accumulated in the environment over a period of years, even though the broad 0–25 yr age-group is, in itself, inhomogeneous with respect to ^{90}Sr concentration,^(13,14) and may account for some of the variability. Since the pooling of two or more log-normal populations with widely different means will not necessarily lead to a log-normal population, the variability of the data of the United States 0–25 age-group might be attributable

* See footnote on p. 1193.

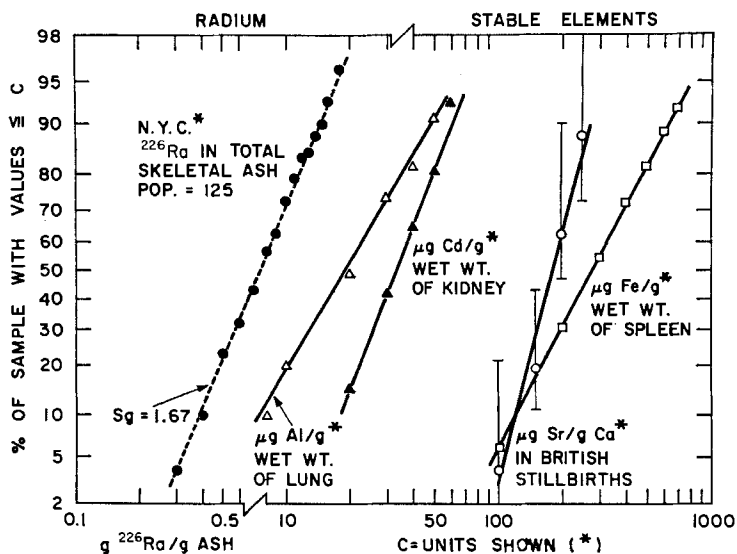


FIG. 3. Cumulative frequency distributions of concentrations of stable elements and radium in human tissues, re-plotted from SYNDER and COOK.⁽²⁵⁾ Vertical "probability" scale is transformed so that distances from the 50% mark are proportional to the number of standard deviations of a standard normal distribution that correspond to the cumulative percentages indicated. The horizontal scale is proportional to the logarithms of the indicated concentrations. The close fit of each plot to a straight line indicates that the concentration data are all consistent with the log-normal shape. ●— ^{226}Ra in total skeletal ash of 125 cadavers, ages 20–91, New York City; △— $\mu\text{g Al/g}$ wet lung from 123 U.S. adults; ages 20–59; ▲— $\mu\text{g Cd/g}$ of wet kidney from 125 U.S. adults, ages 20–59; ○— μg stable strontium per g Ca in 72 British stillbirths, with 95% confidence bands re-plotted from SNYDER and COOK;⁽²⁵⁾ □— $\mu\text{g Fe/g}$ wet weight of spleen from 98 U.S. adults, ages 20–59.

more to individual variations in intake rather than age dependence. However, much larger samples would be needed to subdivide and study the variability of ^{90}Sr concentration within narrower age groups. The shift in the lines for the two different age groups in the Denmark data is consistent with KULP and SCHULERT'S^(13,14) predictions from Langham's model, and with the findings of BRYANT⁽³²⁾ and LOUTIT.⁽³³⁾

In order to determine whether there might be more variability in the distribution of ^{90}Sr per gram bone (which is more closely proportional to radiation dose) than per gram calcium, both the $^{90}\text{Sr/g Ca}$ and $^{90}\text{Sr/g}$ bone data⁽³¹⁾ for the United States (1964–1965) are plotted in Fig. 5. Also, points are plotted on the left-hand part of the abscissa for three separate regions of the United States. These three regions are pooled for the United States as a whole in the plot on the right-hand abscissa using the same size

$^{90}\text{Sr/g Ca}$ and $^{90}\text{Sr/g}$ bone scales for comparison. The data in each of the geographic regions, as well as the United States as a whole, is seen to be approximately log-normally distributed, although in general data *pooled* from two or more log-normal populations (where the log-normal variates themselves are not *added*) will not give a resultant log-normal distribution. In Fig. 5, the data⁽³¹⁾ from the Western part of the United States (stratified only approximately by population density) can be seen to follow a different distribution from those of the Southwest-Central-Delta and Northeast regions. If the Western data is truly log-normal, then the distributions of the pooled U.S. data fit the log-normal so well only as a result of the reduced weighting of the much smaller number of samples from the Western region. However, since the U.S. population is more heavily centered in the East, the distributions of $^{90}\text{Sr/g Ca}$ and $^{90}\text{Sr/g}$ bone given by the lines on the right side of Fig. 5

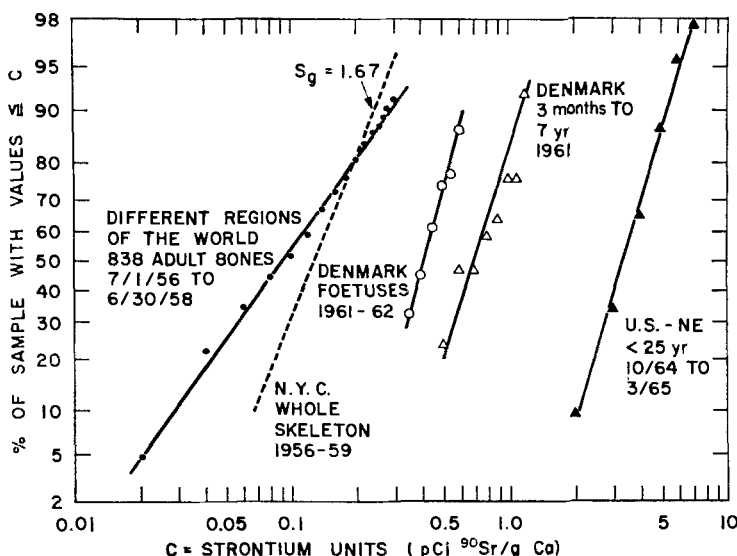


FIG. 4. Comparison of cumulative log-normal plots of $^{90}\text{Sr/g Ca}$ in bone samples from several different populations. ●—human bones of 838 adults collected from different regions of the world, 7/56–6/58;⁽²⁹⁾ --- Kulp and Schultert's fit to whole skeleton samples from New York City, 1956–1959;⁽¹⁴⁾ ●—foetuses from Denmark, 1961–1962;⁽²⁸⁾ △—children 3 months–17 yr old from Denmark, 1961;⁽²⁸⁾ ▲—children and adults under age 25 from Northeastern U.S. 10/64–3/65.⁽³¹⁾

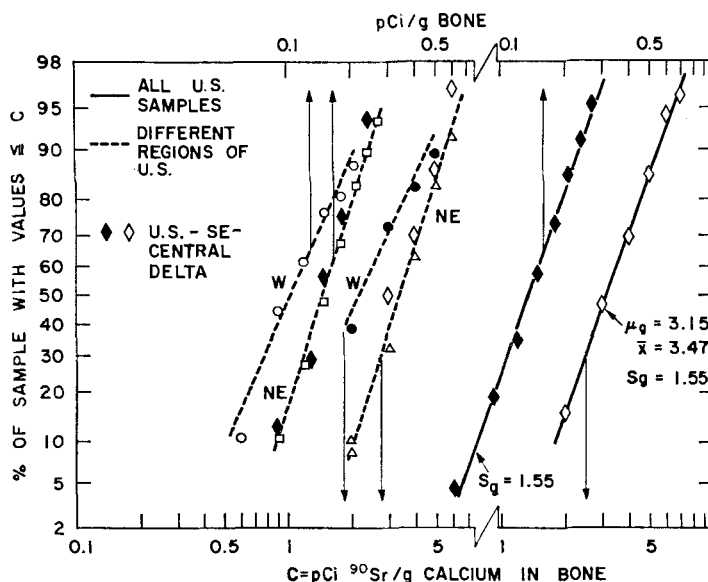


FIG. 5. Comparison of log-normal distributions of $\text{pCi}^{90}\text{Sr/g Ca}$ and $\text{pCi}^{90}\text{Sr/g bone}$ in samples from 224 individuals less than 25 years old throughout the U.S., 10/64–3/65. The dashed lines on the left show the frequency distributions of subdivisions of the population sample into portions from the Northeastern and Western (including Alaska) sections of the United States. Thus, it is apparent that the pooling of two log-normally distributed sub-populations of similar shape can produce an apparent log-normal population if the parameters of the sub-populations are not too different, particularly if the size of one of the sub-populations predominates.

probably provide a good representation of the degree of individual dose variation within the United States population in any given age group. Although variations in radiochemical^(13,30,31) and spectrochemical⁽³⁴⁾ determinations could account for a s_g of about 1.1 in the ^{90}Sr and natural strontium data, most of the variability in these data probably represent true variations in the bone content of strontium between individuals. Since the lines for the whole skeletal radium and strontium/calcium measurements in Figs. 3 and 4 (with $s_g = 1.67$ for both lines) are so closely parallel to the U.S. population ^{90}Sr lines in Fig. 5 (which both have $s_g = 1.55$), very little of the variation in the data of Fig. 5 can be attributed to the variations between different samples of bone from the same individual rather than the actual variations between individuals. Further, since the lines in Fig. 5 for $^{90}\text{Sr}/\text{g Ca}$ and $^{90}\text{Sr}/\text{g bone}$ both have a $s_g = 1.55$, no significant variation in bone dose between individuals may be attributed to variations in bone calcium. Thus, the lines in Fig. 5 should characterize at least an upper limit for the degree of dose variation between individuals in the U.S. population.

Measurements⁽³⁵⁾ of the ^{137}Cs /potassium ratio in 250 people from various parts of the United States also fitted a log-normal distribution, with a $s_g = 1.4$. This somewhat narrower distribution, compared to those for ^{90}Sr , could also be represented fairly well by a normal distribution.

The measurements⁽³⁶⁾ of radium per gram of calcium in fifty bodies from Northwestern United States were log-normally distributed, with $s_g = 1.47$ compared to a $s_g = 1.67$ for the distribution of radium per gram bone ash in New York City skeletons (Fig. 3). Also, the ratio of the median concentration per gram of calcium was 2 times the median concentration per gram ash, similar to the corresponding ratio for the ^{90}Sr data⁽³¹⁾ of Fig. 5. All of the radionuclide and stable nuclide data plotted so far have been consistent with log-normal distributions in concentration between individuals sampled from any given populations.

Radionuclide Concentrations in the Environment

In order to obtain some idea of the major sources of the variability in human bone

concentrations of ^{90}Sr , the concentration frequency distributions of several types of environmental samples were plotted on log-normal probability paper. If a straight line is obtained for the distribution of ^{90}Sr in a particular segment of the food-chain, and the slope of this line is not much steeper than that of the human bone concentrations, then a major part of the variability in human ^{90}Sr can be attributed to the variability in this segment of the food chain.

In Fig. 6, the right-hand plot represents the variability of the ^{90}Sr deposition in the United States in 1963, taken from a map of ^{90}Sr deposition at various points of the world.⁽³⁷⁾ Although only twenty-three points were available from the map, including Alaska and Hawaii, and although the sampling locations may not be exactly stratified according to the areas from which each segment of the population derives its food, it is evident from Fig. 6 that a considerable amount of the variability in human bone concentrations sampled in some manner from the entire United States may be attributed to variations in deposition within various regions of the United States. Part of the variability in deposition may in turn be attributed to variations in rainfall. Of course, it is well-known that there are many additional sources of variation in the ^{90}Sr in food, such as the degree of land cultivation.⁽³⁸⁾ Farm-to-farm differences in ^{90}Sr in milk are also highly correlated with farm-to-farm differences in hay.⁽³⁸⁾ The confidence intervals about the deposition frequencies in Fig. 5, represent the range of $\pm$ one binomial standard deviation ($p - \sqrt{p(1-p)/23}$, $p + \sqrt{p(1-p)/23}$), where p is the percentage frequency indicated on the ordinate divided by 100. Only twenty-three stations were available giving the cumulative ^{90}Sr deposition in the U.S.A. through 1963. The departure of the points from a straight line is not significant; neither can a real deviation from a log-normal distribution in ^{90}Sr per unit area be definitely ruled out.

In the center portion of Fig. 6, the cumulative distributions in average concentrations of ^{90}Sr and ^{137}Cs in milk in May, 1965 for sixty-three sampling stations⁽³⁹⁾ throughout the U.S.A. are plotted on the same coordinate scale for comparison. Although the degrees of variation

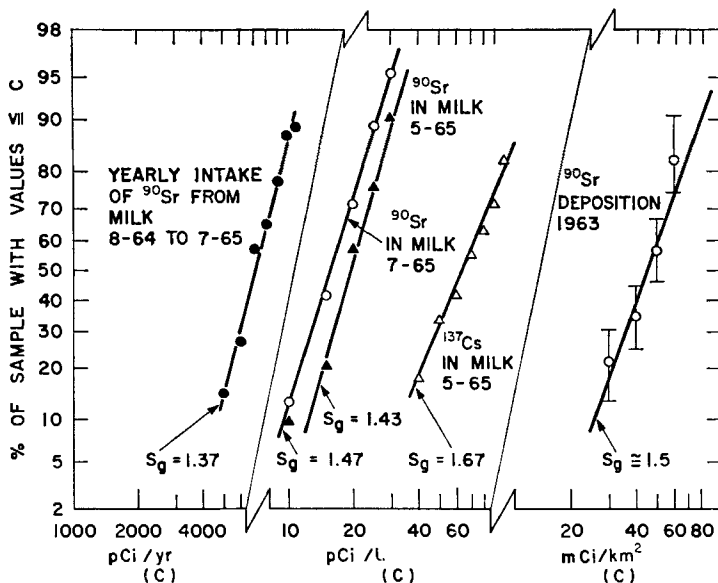


FIG. 6. Comparison of the log-normal frequency distributions of ^{90}Sr deposition and ^{90}Sr and ^{137}Cs concentrations in milk in the U.S. ●—total intake of ^{90}Sr from milk in 1 yr, 8/64–7/65, assuming 1 l./day consumption; ○— ^{90}Sr concentrations in U.S. milk, July, 1965; ▲— ^{90}Sr concentrations in U.S. milk, May, 1965; △— ^{137}Cs concentrations in milk, May, 1965; ◊— ^{90}Sr deposition throughout the U.S., 1963—confidence limits show that distribution of this small sample is statistically consistent with a distribution having a variance similar to that for ^{90}Sr in milk.

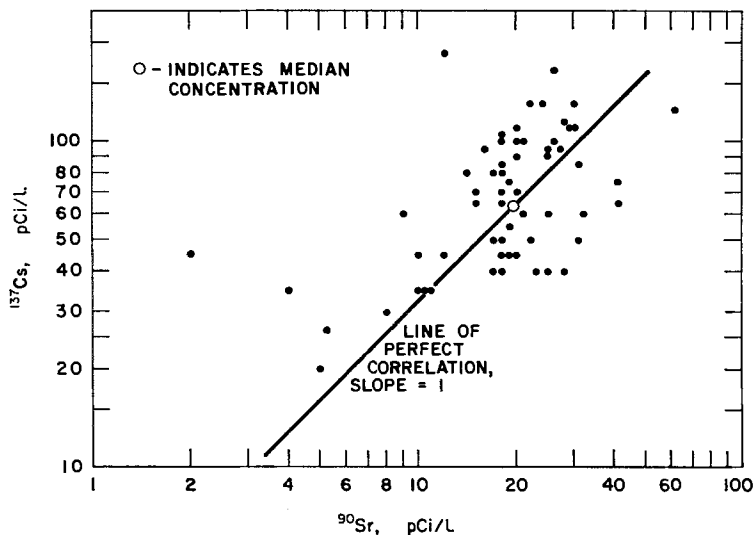


FIG. 7. Scatter diagram showing that ^{137}Cs and ^{90}Sr concentrations of individual milk samples are not well correlated, although the variability of these concentrations throughout all milk samples is similar, for sixty-three stations throughout the U.S. in 1963.

of the ^{90}Sr and ^{137}Cs concentrations are similar, the ^{137}Cs concentration does appear to be significantly more variable. The differences in variability between ^{137}Cs and ^{90}Sr concentrations are probably not attributable to different precisions in the ^{90}Sr and ^{137}Cs analytical procedures. The reported⁽³⁹⁾ standard deviations in the measurements of an individual day's milk sample were only ± 1 pCi/l. for ^{90}Sr and ± 5 pCi/l. for ^{137}Cs . These analytical variations would make relatively minor contributions to the variations in monthly averages about the median concentrations of 20 pCi ^{90}Sr /l. and 64 pCi ^{137}Cs /l., compared to the variations indicated by $s_g = 1.43$ for ^{90}Sr and $s_g = 1.67$ for ^{137}Cs , respectively. The distribution of ^{90}Sr in milk is also plotted in Fig. 6 for July, 1965, in the U.S.A.⁽⁴⁰⁾ The $s_g = 1.47$ is close to that for May, but the line is displaced toward slightly lower values.

In order to determine whether most of the variability in both the ^{90}Sr and ^{137}Cs levels might be attributable to variations in deposition, a scatter diagram (Fig. 7) was plotted of the ^{137}Cs vs. the ^{90}Sr concentration at each station, averaged for May, 1965. Such a diagram may be used as a simple tool for determining the degree of correlation between two variables as another clue (not proof) to possible cause-effect relationships. Although some positive correlation that may depend on the degree of fallout at each station is evident in Fig. 7, the considerable departures of many of the points from the line of perfect correlation show that variations in factors other than deposition, such as calcium and potassium content of the soil, etc., also must play an important role in determining the overall degree of variation in milk concentrations. Thus, the similarity of slopes of these two distributions, which at first may seem to arise from a common cause is found to be fortuitous.

Since the variation in milk concentrations from locality to locality is somewhat less, rather than greater, than the variation in cumulative ^{90}Sr deposition, (Fig. 6) the sampling procedures⁽³⁹⁾ used to obtain daily milk samples at each station probably introduced an averaging process that produced a decrease in variability compensatory to any increase from the independent stochastic variables influencing

transport from soil to milk. However, the milk samples for each of the 63 sampling stations were pooled from each milk processing plant feeding the sampling station in proportion to the plant's average sales in the community,⁽³⁹⁾ so that the frequency distribution in milk concentrations would be of more value in estimating the degree of variation in human internal exposures.

The cumulative frequency distribution in the total annual intake (from August, 1964 through July, 1965) of ^{90}Sr from milk by children at different localities in the United States, assuming an intake of 1 l./day for each child⁽⁴⁰⁾ is illustrated on the left side of Fig. 6. This frequency distribution, for the same sixty-three sampling stations plotted for the May, 1963 data, also is consistent with a log-normal function. For this distribution, $s_g = 1.37$, which indicates that about 68% of the intakes fall within 1.37 and 1/1.37 times the median intake of about 7000 pCi. Although this s_g is based on 19 monthly averages, it is only slightly less than the $s_g = 1.43$ obtained by averaging at each sampling station over the 1-month May 1963 period (Fig. 6). Thus, an averaging period at each location of only 1 month may suffice to obtain a representative picture of the distribution in milk levels throughout the United States. Several other graphs of frequency distributions of ^{90}Sr in food samples also were consistent with the log-normal function and were within the degrees of variability shown by the milk concentrations in Fig. 6.

In order to illustrate some possible applications of the log-normal model in studying the transport of ^{90}Sr through the biosphere, let us assume that

- (a) The ratio R of ^{90}Sr /calcium in new bone formed during 1964–1965 is given by the equation

$$R = f \times l / (\text{annual calcium intake per child}), \quad (15)$$

where f is the fraction of the annual amount of ^{90}Sr ingested that is deposited in bone and l is the total annual intake in picocuries;

- (b) the variation between children in calcium intake is negligible compared to the variation in intake of ^{90}Sr between localities of the United States;

- (c) the variations in R and l are each determined to be log-normally distributed.

Then, assuming that the co-variance of l and f is zero, the degree of variation in f may be obtained by the relationship (see equations (5)–(7)).

$$(\ln s_g)_R^2 = (\ln s_g)_f^2 + (\ln s_g)_l^2. \quad (16)$$

If we take s_g for the intake l as 1.37 from Fig. 6, and s_g for R as 1.55 from Fig. 5, then we might estimate the standard geometric deviation s_g for f as follows:

$$\begin{aligned} (\ln s_g)_f^2 &= (\ln 1.55)^2 - (\ln 1.37)^2 \\ &= (0.438)^2 - (0.315)^2 = 0.093 \end{aligned} \quad (17)$$

$$(\ln s_g)_f = 0.305 \quad (18)$$

$$s_g = 1.36. \quad (19)$$

This would indicate, if the assumptions above were validated, that the variability in human uptake factors f were such that about 68% of the population would have uptake factors f lying between 1.36 and $1/1.36$ times the median; and 95% would lie between $(1.36)^2 = 1.85$ and $1/1.85$ times the median.

It may also be of interest that a re-plot of the frequency distribution given by BECK⁽⁴⁹⁾ of external natural radiation dose-rates at 210 different points throughout the United States in 1962–1963 fitted a log-normal distribution with a median of $5.5 \mu\text{R/hr}$ and a standard geometric deviation $s_g = 1.67$ —the same as the s_g for ^{137}Cs levels in milk. This agreement between the variability of ^{137}Cs in milk and natural external radiation levels may be partly or wholly fortuitous since, as Beck points out, the sampling stations were not necessarily “representative”, or selected in proportion to population density.⁽⁴⁹⁾ However, the similarity in the variability of ^{137}Cs in milk and the external natural radiation levels (which include a substantial contribution from “natural fallout” of ^{222}Rn decay products⁽⁴⁹⁾) may be somewhat suggestive of a log-normal frequency distribution in external population exposures. Moreover, the shape and width ($s_g = 1.67$) of the distribution in natural external radiation levels is about the same as that for the distribution in concentrations of uranium in granite.⁽⁵⁰⁾ In

view of the fact that practically all natural-occurring elements including thorium⁽⁵⁰⁾ are log-normally distributed,⁽⁸⁾ it is reasonable to expect that natural external radiation levels might be log-normally distributed. The similar log-normal shape and s_g for ^{137}Cs in milk and for the frequency distributions of fallout and other materials in the environment may be “fortuitous”, in the sense that the authors cannot prove a common origin for both. However, the shapes and the values of s_g are so close that by any statistical test with ordinary probability levels of acceptance they obviously could be considered consistent with a common underlying frequency distribution; i.e. they are indeed extremely “fortuitous”. Thus, these observations are reported for whatever possible leads they may give to other investigators, just in case the authors are blind to the existence of some highly probable relationships.

The frequency distribution of biological half-lives of tritium as replotted from BUTLER and LEROY⁽⁵¹⁾ also closely approximates a log-normal distribution. Thus, a further examination of the various parameters affecting exposure from fallout nuclides may indicate that they can generally be characterized by log-normal distributions for purposes of determining the magnitudes and sources of variability in population exposures.

IV. DISCUSSION

The precise estimation of exposure distributions to various individual nations or the world population would require a division of these populations into sub-groups and stratified sampling of each sub-group according to population size (or according to population feeding from a given farm production area in the case of food sampling). Nevertheless, sufficient information is now available from various sampling locations to pinpoint the major trends. Also, ^{90}Sr and other nuclide concentrations from samples collected from the United States and other countries within the past few years have been repeatedly fitted to log-normal distributions having geometric standard deviations within a fairly narrow range. Thus, the properties of the log-normal distribution may be used to estimate average population exposure from ^{90}Sr in the United States, and the

variations in exposure between members of the population over a considerable range of the exposure frequency distribution.

Table 1 summarizes the estimated percentages of the United States population above various ^{90}Sr concentrations and annual dose levels, for the 1964–1965 data⁽³¹⁾ as plotted in Fig. 5. The annual dose estimates in Table 1 were obtained from the ^{90}Sr concentrations in pCi/g Ca, using conversion factors suggested by the U.S. Federal Radiation Council.⁽⁴¹⁾ These dose conversion factors were intended for the higher-exposed infant and fetal populations,⁽⁴¹⁾ who have the higher ^{90}Sr concentrations.^(13,14,30,37) Dose conversion factors for adults would be higher by a factor of three or more, but bone concentrations would be considerably lower.^(13,37)

If the 1959 ICRP-NCRP maximum permissible bone burden⁽²¹⁾ of about 2 μCi in standard man (having 1000 g Ca in his bone) were assumed to produce the maximum permissible occupational dose rate of (0.56 rem/wk)/5 rad/week,⁽²¹⁾ then 1 pCi/1000 g Ca would produce an annual dose to mineral bone of about

$$\frac{(0.56/5) \times 52 \text{ rad/year}}{2 \mu\text{Ci} \times 10^6 \text{ pCi}/\mu\text{Ci}} = 2.9 \times 10^{-6} \text{ rad/year}$$

neglecting the low elimination rate of ^{90}Sr from bone. Thus, 1 pCi/g Ca would produce a dose-rate of 2.9 mrad/yr to the mineral bone of adults. The dose conversion factor for mineral bone in infants and children has also been assumed to be about this magnitude.⁽⁴¹⁾ Thus, it can be seen from Table 1 and previous data^(13,15) on bone concentrations as a function

of age, (see also Fig. 4), that either the marrow or mineral bone doses of most of the U.S. adult or child population will be small compared to natural background exposures.⁽³⁷⁾

The bone doses will also be small compared to the British⁽⁴²⁾ action guide of 1.5 rad/yr local dose to mineral bone (which may be perhaps 10 times the average), or the United States⁽⁴³⁾ "Category I" action guide level of 10 rads average dose to bone marrow and 30 rads to mineral bone for the first year, for which protective action should be considered following a contaminating incident.⁽⁴⁴⁾ However, the United States Protective Action Guides were not intended to apply to long-term fallout, for reasons given in Ref. (43). Instead it is stated that "... annual radiation doses from fallout equal to or greater than the numerical values of the RPG's can be used as an indication of when there is a need to initiate a careful evaluation of fallout exposures."⁽⁴³⁾ The U.S. Radiation Protection Guide (RPG) for peacetime exposure of the general population from man-made sources is 0.5 rem/yr for individual members of the population and 0.17 rem/yr for the average of a "suitable" sample.⁽⁴³⁾ Table 4 indicates that the marrow dose for the more sensitive, higher-exposed, 0–25 age group averages only 3.1 mrad/yr for this age group, and only about 0.36 % of this age group exceeded 9.3 mrad/yr in 1964–1965. The adult population will have much lower exposures with about the same degree of variability (see Fig. 4). Thus, the ^{90}Sr exposures in humans are still relatively low compared to the United States RPG's.

Table 1. Estimates of percentages of United States 0–25 age group above various ^{90}Sr levels, 1964–1965

^{90}Sr level	^{90}Sr concentration *		^{90}Sr bone marrow dose- (mrad/yr)	Percent of U.S. in 0–25 age with ^{90}Sr levels greater than indicated *
	(pCi/g Ca)	(pCi/g bone)		
Median	3.15	1.35	2.8	50
Average	3.47	1.49	3.1	40
2 $\times$ Average	6.94	2.70	6.2	3.6
3 $\times$ Average	10.41	4.05	9.3	0.36

* From data of Ref. 31 as plotted in Fig. 5.

† Using the conversion factor of Ref. 41 that (for fetus or infant) 1 nCi of ^{90}Sr /g Ca in the skeleton will give an average bone marrow dose of 0.9 rad in 1 yr.

The average concentration of 3.47 pCi/g Ca given in Table 1, as obtained from Fig. 5, compares favorably with the peak equilibrium level of $3.5 \mu\text{Ci } ^{90}\text{Sr/g Ca}$ predicted by KULP and SCHULERT^(13,14) for children and young adults at 4–5 yr following the 1961 Soviet tests. It is also consistent with the peak levels of 40 pCi/g Ca found in milk in 1964,⁽⁴³⁾ (see also Fig. 6) the milk-human discrimination factors, and the proportion of ^{90}Sr ingested from milk.^(13,14,15,45) The 1964–1965 ^{90}Sr in bone concentrations from Fig. 5 for the 0–25 yr age group are also in the same order of magnitude as the 1.9 pCi $^{90}\text{Sr/g Ca}$ and 5.0 pCi $^{90}\text{Sr/g Ca}$ levels in 0.4-yr-old bone in “dry” and “wet” areas of the United States, respectively, in 1963,⁽⁴⁶⁾ indicating that the plateau⁽¹³⁾ in ^{90}Sr in young children has probably been reached, provided that no significant amount of further atmospheric weapon-testing occurs.

Although additional population radiation exposure from fallout occurs from internal sources other than ^{90}Sr as well as external sources, the total average annual exposure for children and young adults should not exceed about twice that from ^{90}Sr alone, except perhaps for fetuses,⁽⁴⁷⁾ in which the ^{90}Sr levels are relatively low^(13–15) (see Fig. 4). On the other hand, it should be kept in mind that bone doses in “hotspots” of newly-formed bone may be up to about 10 times higher⁽⁴⁵⁾ than the bone doses averaged over the skeletal mass, as given in Table 1. Nevertheless, the lowest experimental average bone doses from ^{90}Sr at which experimental leukemogenesis has been induced are about 900 rad over the animal lifespan,⁽⁴⁵⁾ so that the possible cancer incidence at ^{90}Sr doses on the order of 0.01–0.1 rad/yr is unknown but believed to be small. Bone cancer has been found so far to occur generally at higher experimental doses than leukemia.⁽⁴⁵⁾ Epidemiologic data on leukemogenesis in man has produced estimates of an incidence on the order of $1 - 4 \times 10^{-6}$ cases per person-rad-year under observation,⁽¹⁵⁾ but these data have been obtained at dose rates many orders of magnitude greater than 0.1 rad/yr. Limited experimental data in recent years has indicated that lower dose rates will produce a lower leukemia incidence per person-rad-year.⁽⁴⁵⁾ Also, radiation induced tumors other than leukemia have

been found to occur at lower incidences than leukemia.⁽⁴⁵⁾

Although some of the earlier bone data from worldwide samples indicated relatively wide frequency distributions in ^{90}Sr concentration,^(3–5) the more recent data give narrow distributions within limited geographic regions (Figs. 3–5). There is in fact some indication in Figs. 3–5 that ^{90}Sr in newly-formed bone reaches an equilibrium degree of variability between individuals of the same age group after the fallout, weather, and ecological patterns of a given region has been averaged (or cumulated) over several years or more. Since the Northern latitudes in which the majority of the United States population resides receive the largest fallout deposits and consume the most ^{90}Sr ,^(13,15,48) the fraction of the total U.S. population and of the world population that in 1964–1965 exceeded the level of 10.4 pCi/g Ca (Table 1) should be much smaller (perhaps an order of magnitude or more smaller) than the 0.36% indicated for the United States 0–25 age group, if the variability of concentrations within each country is assumed to be no greater than that shown in Fig. 5 for the United States.

Thus, although some of the apparent log-normal distributions in worldwide bone concentrations would predict much larger percentages of the world population exceeding given multiples of the world average, the world average would be much lower than the United States average for the 0–25 age group. Further, the extrapolation of an apparent log-normal line for the world population to any higher concentrations beyond the range of the data may not be valid since the slope of the world frequency distribution may be expected to increase rapidly as the upper percentiles are approached; i.e. after the majority of persons in the Northern Latitudes have been accounted for (Figs. 3 and 5). Moreover, these highest concentrations occur in sub-populations representing a very small fraction of the world population.^(13,43,46)

In order to ultimately improve the estimates of the exposure distribution to the world population from fallout, the sampling network within each country should be divided into a number of areas of about equal population size. Bone samples should be collected from each age

group within each area in proportion to the population size of the age group within the area—not in proportion to the death rate of the age group. Each sub-population should also be divided according to other characteristics that may influence ^{90}Sr uptake and retention, such as calcium level in the diet, etc., depending on the purpose of the study and data available. Results for each individual sample within each age-area sub-group should be identified and reported with that sub-group, and should not be pooled prematurely for the area as a whole. With appropriately designed sampling and calibration procedures, and with analyses conducted in such a manner that data can be compared, on an absolute basis, over a long period of time, a better evaluation of the distribution of ^{90}Sr throughout the world population could be obtained. If log-normal distributions were then found to characterize the variability within all or most age-area sub-groups, simple computing procedures could be devised to provide a more precise and continuing estimation of fallout dose distributions for ultimate use in estimating or predicting chronic effects (if any) of low-level irradiation of large populations by fallout radionuclides. Statistical procedures for the maximum-likelihood estimation of the parameters of a log-normal distribution from a given set of data are already available.⁽¹²⁾ Analogous procedures may also be applied to the estimation of population exposures from external irradiation by fallout radionuclides on the ground.^(15,49)

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